

Additive and Subtractive Modes of Trivista

The TriVista spectrometer can be used in single, double and triple configurations. Single configuration (Fig 1A) means all three stages can be used simultaneously and independently for three different projects. This is highly practical but quite rare situation. The most often TriVista is utilized as a double or triple system (Fig 1B,C) when light beam is passed sequentially through 2 or 3 stages and gratings of the involved stages coherently move together with very high precision. Two most common reasons why people use double or triple system instead of a single spectrometer are *high spectral resolution* and *high stray light rejection*. These two effects can be achieved in different modes of TriVista operation:

1. Additive mode gives high spectral resolution and high linear dispersion.
2. Subtractive mode gives high stray light rejection.

Additive mode

In this mode gratings on all 3 stages of spectrometer contribute to positive light dispersion (Fig.2A). Polychromatic light enters the first stage of the instrument through the slit S_1 . Grating G_1 disperses the light. Slit $S_{1,2}$ acts as a bandpass filter. It passes through only narrow portion of spectrum which is further dispersed on the grating G_2 . Light passing through the slit $S_{2,3}$ is dispersed for a third time on the grating G_3 and is projected on detector D.

The dispersion of TriVista in additive mode is defined by adding dispersions of each stage. If all three stages have the same gratings and focal lengths the dispersion is simply equals to triple dispersion of its any single stage. Let us consider TriVista555 as an example where focal length for all three stages equal 500 mm and 1200 g/mm gratings are used on each stage. Single 500 mm stage with 1200 g/mm grating has a nominal linear dispersion of 1.7 nm/mm. In the triple additive mode linear dispersion of TriVista555 is approximately equal to 0.5 nm/mm for the center of visible spectral range. While working with CCD detector in additive mode slits $S_{1,2}$ and $S_{2,3}$ have to be opened relatively wide to allow spectral coverage of the whole CCD. Broad slit width inevitably

contributes to excessive stray light. When working with PMT slit $S_{2,3}$ can be kept as narrow as 10 mkm. In this case high resolution and high stray light rejection can be achieved simultaneously. However, the disadvantage of working with PMT is a necessity for time consuming scanning in order to do measurements in a wide spectrum range.

Subtractive mode

Excellent stray light rejection with CCD detector can be achieved when subtractive mode is used. In this mode the first and the second stages of spectrometer work as a tunable bandpass filter cutting out a desirable portion of spectrum and projecting it on CCD (Fig. 2B). A polychromatic light enters the first stage through the entrance slit S_1 and is dispersed by the grating G_1 . Slit $S_{1,2}$ again acts as a bandpass filter passing only the light between wavelengths λ_1 and λ_2 . The grating G_2 recombines all the dispersed light and focuses it into the middle of the slit $S_{2,3}$ producing again a polychromatic light limited to the spectral range between wavelengths λ_1 and λ_2 . Grating G_3 disperses this light again and projects it into detector. Slit $S_{1,2}$ in subtractive mode is usually relatively wide opened to allow desirable spectral range λ_1 - λ_2 to pass through. But the slit $S_{2,3}$ is normally very narrow which assures high stray light rejection.

Gratings G_1 and G_2 in subtractive mode must match in groove density and therefore their dispersion actions totally cancel each other. They act as a very sharp bandpass filter allowing for Raman measurements very close to the laser line. In this sense spectral resolution of TriVista is entirely defined by the spectral resolution of the third stage depending on slit $S_{2,3}$ width, grating G_3 groove density and the third stage focal length.

It is possible to use CCD or PMT detector with TriVista working in subtractive mode. Though, CCD type is preferable because of reduced time required to take measurements.

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Easy Switch between Additive and Subtractive modes

TriVista software offers an Easy Switch between additive and subtractive modes just by mouse-clicking. The physical mechanism behind this switch is changing direction of gratings rotation. In additive mode both gratings S_1 and S_2 synchronously rotate clockwise adding dispersion to each other. In subtractive mode grating S_1 rotates clockwise but the grating S_2 rotates counter-clockwise precisely canceling dispersive action of the grating S_1 .

Modes	Double Additive	Double Subtractive	Triple Additive	Triple Subtractive
Primary effect	High resolution	High stray light rejection	Best resolution	Best stray light rejection
Sec. effect	High stray light rejection (with PMT and narrow slit S2,3)		High stray light rejection (with PMT and narrow slit S2,3)	
CCD	Y	N	Y	Y
PMT	Y	Y	Y	Y

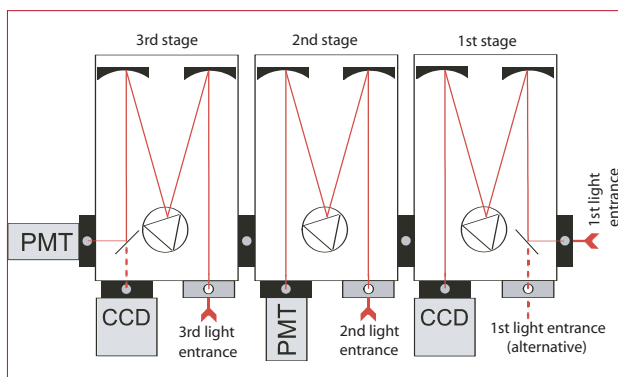


Figure 1A. Single Configuration

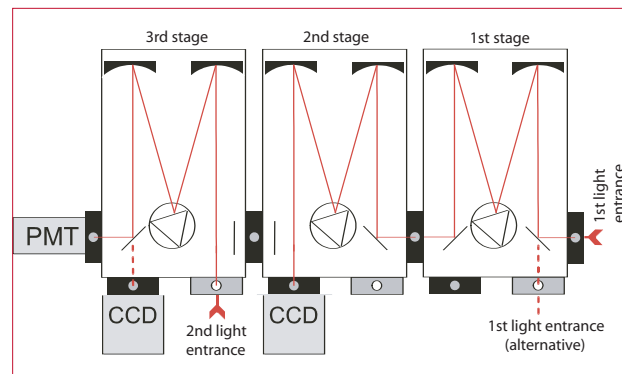


Figure 1B. Triple configuration

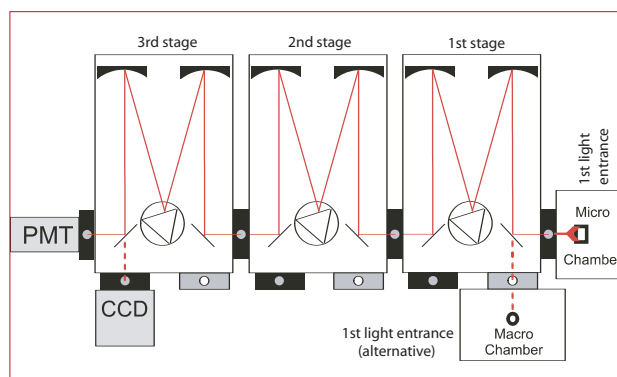


Figure 1C. Double Configuration

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PIACTON

www.piacton.com

email: moreinfo@piacton.com

USA +1.609.587.9797 | Benelux +31 (347) 324989

France +33 (1) 60.86.03.65 | Germany +49 (0) 89.660.779.3

Japan +81.3.5639.2741 | UK +44 (0) 28.38310171

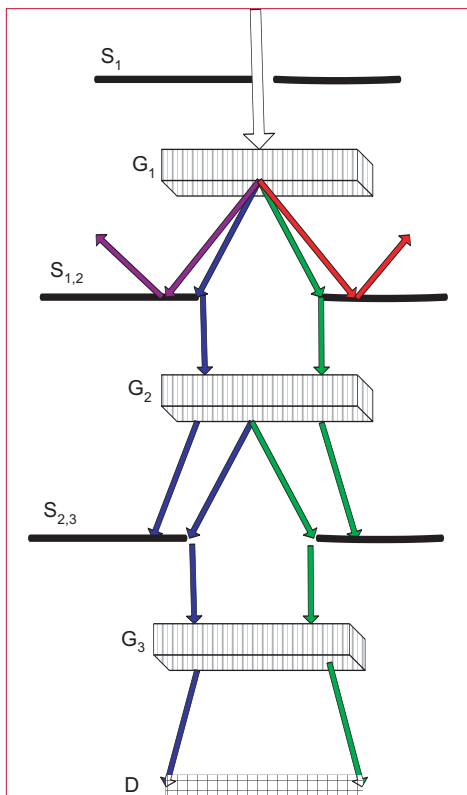


Figure 2A. Additive mode

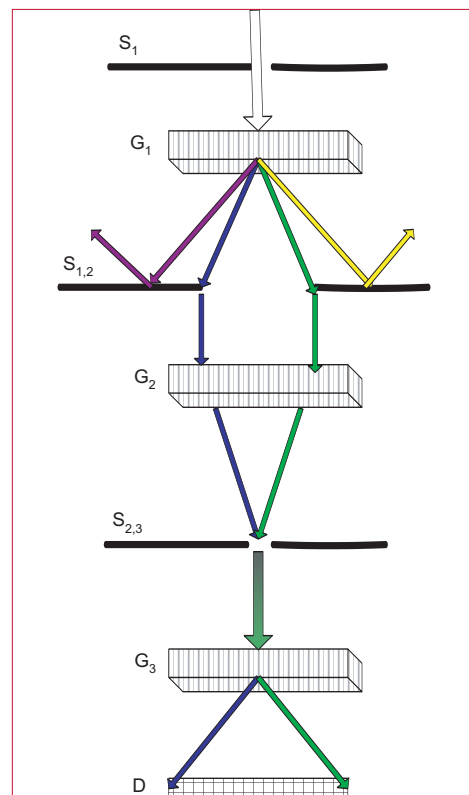


Figure 2B. Subtractive mode

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PI•ACTON

www.piacton.com

email: moreinfo@piacton.com

USA +1.609.587.9797 | Benelux +31 (347) 324989

France +33 (1) 60.86.03.65 | Germany +49 (0) 89.660.779.3

Japan +81.3.5639.2741 | UK +44 (0) 28.38310171